Semester 5 · Program Core · 1.5 Credits · Practical course

Machine Shop Practice

Practical handbook for production type lathe operations, milling, gear cutting, drilling, planing, slotting, shaping and grinding operations with safety, inspection and lab-exam preparation.

Course purpose

This course turns machine-tool theory into workshop skill. Students must set jobs, choose tools, perform operations, measure output, maintain safety and finish the required practice models accurately.

Malayalam: Machine shop practice-□ operation □□□□□□□□□□□ □□□□□□□ □□□□. Job holding, tool setting, feed/speed, measurement, safety, finishing □□□□□□□ □□□□□□□□ □□□□□□□□.

Official details

Code: 5027 · **Title:** Machine Shop Practice · **Program:** Mechanical / Manufacturing · **Periods:** 45 · **Prerequisite:** Workshop practice 3 and 4.

CO1

Production lathe operations.

CO2

Milling, gear cutting and drilling.

CO3

Planing and slotting operations.

CO4

Cylindrical, surface and centerless grinding.

Practical roadmap

CO1 · 10 Hours

Production Type Lathe Operations

Taper and form turning

Taper turning produces conical surfaces by setting tool or work motion at an angle. Form turning uses a form tool to reproduce a profile on the work. Check job holding, tool height, feed and cutting speed before starting.

Thread cutting

Thread cutting needs correct gear/feed setting, tool angle, compound slide setting, trial cut and pitch verification. Use cutting fluid and stop before shoulder safely.

Production lathe practice

Hexagonal or square headed bolts are produced by feeding square or hex bars in production, heavy duty, capstan or turret lathes.

Checklist

1. Confirm drawing and dimensions.

2. Check chuck/collet grip.
3. Set tool on centre height.
4. Measure after roughing and finishing.

CO2 · 10 Hours + Lab Exam

Milling, Gear Cutting and Drilling

Milling and indexing

Milling exercises include making square and hexagon from round bars using simple/direct, compound and differential indexing methods. Correct indexing and clamping avoid unequal faces.

Gear cutting

Spur and helical gear teeth generation requires blank preparation, cutter selection, indexing, depth setting and repeated tooth spacing. Wrong index calculation will scrap the job.

Drilling exercise

Drilling practice includes three different sized holes on different materials with uniform distances. Marking, centre punching, pilot drilling and deburring are essential.

Expected viva

1. What is indexing?
2. Differentiate simple and differential indexing.
3. Why pilot hole is used?
4. List drilling safety precautions.

CO3 · 12 Hours

Planing, Slotting and Shaping Practice

Planing and T-slots

Planing produces flat surfaces on large workpieces. T-slot exercise requires rough slot, undercutting and final measurement. Set stroke length and avoid tool collision.

Slotting and keyways

Slotting produces internal keyways, grooves and splines. The tool reciprocates vertically. Correct job setting and slow feed are important for straight slots.

Shaping models

Practice includes shaping a hexagon on a round bar and cutting step block dovetails at 60, 90 and 120 degrees.

Safety

Keep hands away from ram path. Remove chips with brush. Clamp job rigidly. Do not measure while machine is running.

CO4 · 10 Hours + Lab Exam

Grinding Operations

Cylindrical grinding

Cylindrical grinding finishes external and internal circular surfaces. It requires correct wheel selection, dressing, job support, coolant and small depth of cut.

Surface and centerless grinding

Surface grinding produces flat accurate surfaces. Centerless grinding supports work between grinding wheel, regulating wheel and work rest.

Tool grinding

Cutting tools are ground to required angles. Maintain rake, clearance and cutting edge geometry. Overheating destroys tool hardness.

Grinding safety

1. Ring-test wheel before mounting.
2. Use guard and goggles.
3. Do not exceed wheel speed.
4. Use light cuts and coolant where required.

Workshop rule bank

Measure twice, cut once. Never wear loose clothing near rotating machines. Stop machine before measuring. Use brush for chips. Keep tool sharp and work clamped.

Lab model preparation

1. Read drawing and tolerance.
2. Select machine, work holding and tools.
3. Set datum and perform roughing.
4. Measure and finish.
5. Deburr and submit inspection report.

Answer guide

For practical records include objective, tool list, machine used, drawing, procedure, safety precautions, observation table, result and viva answers. For lab exam, accuracy and safe method are more important than speed.

References: Myron Begman, HMT Production Technology, Hajra Choudhury and Bhattacharya, R.K. Jain, P.C. Sharma, Taylor and Francis, Elsevier, ScienceDirect and SAE resources.